

Node.js - Programs

```
Last login: Wed Sep 16 09:07:47 on ttys010
[amrutha@Amruthas-MacBook-Air ~ % node --version
v26.5.1
[amrutha@Amruthas-MacBook-Air ~ % npm --version
11.17.0
```

Fig 1: Verification of Node.js and npm installation on macOS.

Create the first Node.js program

```
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console.log("Hello Students!");
console.log("Welcome to Node.js");
```

Fig 2: Basic Node.js program using app.js to display messages in the terminal.

```
[amrutha@Amruthas-MacBook-Air ~ % mkdir NodeJS_Practical
[amrutha@Amruthas-MacBook-Air ~ % cd NodeJS_Practical
[amrutha@Amruthas-MacBook-Air NodeJS_Practical % nano app.js
[amrutha@Amruthas-MacBook-Air NodeJS_Practical % node app.js
Hello Students!
Welcome to Node.js
amrutha@Amruthas-MacBook-Air NodeJS_Practical %
```

Fig 3: Execution of the basic Node.js program using app.js.

```
http = require('node:http');

listener = function (request, response) {
  response.writeHead(200, {'Content-Type': 'text/html'});
  response.end('<h2 style="text-align: center;">Hello World</h2>');
};

server = http.createServer(listener);
server.listen(3000);

console.log('Server running at http://127.0.0.1:3000/');
```

Fig 4: Node.js HTTP server program using the http module in hello.js.

```
amrutha@Amruthas-MacBook-Air NodeJS_Practical % nano hello.js
amrutha@Amruthas-MacBook-Air NodeJS_Practical % node hello.js
Server running at http://127.0.0.1:3000/
```

Fig 5: Execution of the Node.js HTTP server using hello.js.



Fig 6: Browser output of the Node.js HTTP server displaying “Hello World”.

```
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const http = require("http");

const server = http.createServer((req, res) => {
  res.writeHead(200, {
    "Content-Type": "text/plain"
  });

  res.end("Hello Students! Node.js Server is Running.");
});

server.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

Fig 7: Node.js HTTP server program using server.js.

```
amrutha@Amruthas-MacBook-Air NodeJS_Practical % nano server.js
amrutha@Amruthas-MacBook-Air NodeJS_Practical % node server.js
Server running at http://localhost:3000
```

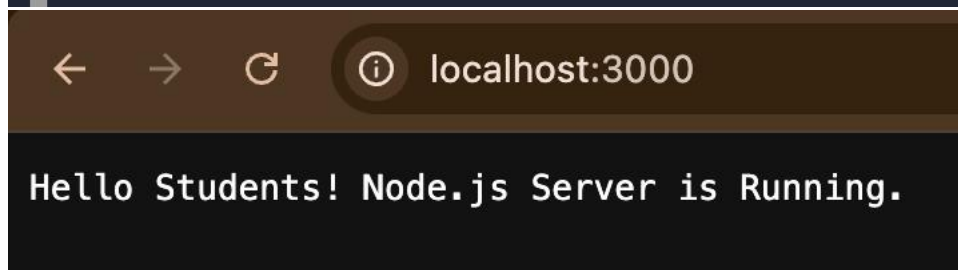


Fig 8 & 9: Execution of the Node.js HTTP server on port 3000 and Browser output of the Node.js HTTP server.

```
[amrutha@Amruthas-MacBook-Air NodeJS_Practical % npm init -y
Wrote to /Users/amrutha/NodeJS_Practical/package.json:

{
  "name": "nodejs_practical",
  "version": "1.0.0",
  "description": "",
  "main": "app.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Fig 10: Creation of the package.json file using the npm init -y command.

```
[amrutha@Amruthas-MacBook-Air NodeJS_Practical % node
Welcome to Node.js v26.5.1.
Type ".help" for more information.
[> 10+2
12
[> a=10
10
[> b=20
20
[> a+b
30
> █
```

Fig 11: Execution of simple expressions and variable operations in the Node.js console.

```
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// Variable store number data type
let a = 35;
console.log(typeof a);

// Variable store string data type
a = "GeeksforGeeks";
console.log(typeof a);

// Variable store Boolean data type
a = true;
console.log(typeof a);

// Variable store undefined (no value) data type
a = undefined;
console.log(typeof a);█
```

Fig 12: Node.js program demonstrating different JavaScript data types using typeof.

```
amrutha@Amruthas-MacBook-Air NodeJS_Practical % nano console.js
amrutha@Amruthas-MacBook-Air NodeJS_Practical % node console.js
number
string
boolean
undefined
amrutha@Amruthas-MacBook-Air NodeJS_Practical %
```

Fig 13: Output of the Node.js data type program.

```
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console.log("Hello World");
```

Fig 14: Simple Node.js program to display “Hello World” using console.log().

```
amrutha@Amruthas-MacBook-Air NodeJS_Practical % node hello_console.js
Hello World
amrutha@Amruthas-MacBook-Air NodeJS_Practical %
```

Fig 15: Output of the simple “Hello World” Node.js program.